MATH 165

COLLEGE ALGEBRA WITH ANALYTIC GEOMETRY

E-PORTFOLIO

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Academic Year 2026–2027

COURSE OUTCOME 1

CW1.1-WCR

CW1.1-WCR

100%

Last sync a month ago

15 questions

75 / 75 points

Correct.

Solve for x.

$$1-\frac{x-5}{5}=\frac{x+3}{4}-\frac{3x-3}{20}$$

Enter the exact answer as a fraction. Use "/" as the division sign. Use an improper fraction if necessary, do not use a mixed fraction.

$x =$ 11/3

Correct.

Solve the radical equation for x.

$$\sqrt{4-x}=-2+\sqrt{2x+3}$$

If there is no solution, enter "NA".

$x =$ 3

CW1.2-WCR

CW1.2-WCR

100%

Last sync a month ago

15 questions

75 / 75 points

Correct.

Please view the following video before answering this question. Lecture Video, Section 6.1, Example 01
Use the substitution method to solve the following system of linear equations:

$$\begin{cases} 2x - y = -5 \\ 4x + y = 2 \end{cases}$$

(-1/2 , 4)

Correct.

Please view the following video before answering this question. Lecture Video, Section 6.1, Example 05
Use the elimination method to solve the following system of linear equations:

$$\begin{cases} 2x + y = 11 \\ x + 3y = 18 \end{cases}$$

If there is no solution, enter "none" in both entry areas.

(3 , 5)

CW1.3-WCR

CW1.3-WCR

100%

Syncing error. Try again.

7 questions

65 / 65 points

Correct.

Please view the following video before answering this question. Lecture Video, Section 8.5, Example 05
Solve the system of equations and graph to verify the answer.

$$x^2 + y^2 = 4$$
$$y = x^2 - 9$$

- $\left(-\sqrt{3}+1,\sqrt{2}\right), \left(-\sqrt{3}+1,\sqrt{2}\right)$
- No solution
- $\left(1,-\sqrt{3}\right), \left(1,\sqrt{3}\right)$
- $(2,0), (-2,0)$
- $(0,2), (2,0)$

✔ Correct.

Please view the following video before answering this question. Lecture Video, Section 8.5, Example 05
Solve the system of equations and graph to verify the answer.

$$\begin{aligned} y &= 2x + 1 \\ 2x^2 + y^2 &= 1 \end{aligned}$$

- $(0,1), \left(-\frac{1}{3}, -\frac{2}{3}\right)$
- $\left(-\frac{2}{3}, -\frac{1}{3}\right), \left(-\frac{1}{3}, -\frac{2}{3}\right)$
- $\left(\frac{\sqrt{6}}{6}, \sqrt{\frac{2}{3}}\right), \left(-\frac{\sqrt{6}}{6}, -\sqrt{\frac{2}{3}}\right)$
- $\left(-\frac{1}{3}, -\frac{1}{3}\right), (0,1)$
- $(0,1), \left(-\frac{2}{3}, -\frac{1}{3}\right)$

CW1.4-WCR

CW1.4-WCR

100%

Syncing error. Try again.

10 questions

50 / 50 points

Your answer is correct.

Solve the polynomial inequality and express the solution set in interval notation.

$$t^2 < 16$$

The format of the interval is (,) ✓

(−4 ✓ , 4 ✓)

✔ Your answer is correct.

Solve the polynomial inequality and express the solution set in interval notation.

$$x^2 - 2x > 2$$

NOTE: Enter the exact answer.

The format of the interval is (,) ∪ (,) ✔

Interval = $\left(\boxed{-\infty}, \boxed{1-\sqrt{3}}\right) \cup \left(\boxed{1+\sqrt{3}}, \boxed{\infty}\right)$

✔ ✔ ✔ ✔

CW1.5-WCR

CW1.5-WCR

100%

Syncing error. Try again.

9 questions

45 / 45 points

Correct.

Please view the following video before answering this question. Lecture Video, Section 6.5, Example 2

Find the minimum value of $z = x + 2y$ subject to the constraints.

$$y \leq -x + 5$$

$$x \geq 0$$

$$y \geq 0$$

$z =$ 0

Correct.

Minimize $z = 3x + 9y$ subject to:

$$x \geq 0 \qquad y \geq 0$$

$$x + y \leq 8 \qquad x + y \geq 0$$

The minimum value of z is 0 , which occurs at the point (0 , 0).

CW1.1-CWA

Tutorial Exercise

Solve the absolute value equation for x.

$$|x - 6| = 5$$

Step 1

The property of absolute value equations states that for any variable expression E and any nonnegative real number k, $|E| = k$ if and only if $E = k$, or

$E =$ $-k$ ✓ .

Step 2

We are asked to solve the absolute value equation $|x - 6| = 5$ for x. Thus, $|x - 6| = 5$ implies

$x - 6 =$ 5 ✓ or $x - 6 =$ -5 ✓ . (Enter the larger answer first.)

Step 3

Solve each equation for x.

$x - 6 = 5$

$x =$ 5+6 ✓

or

$x - 6 = -5$

$x =$ -5+6 ✓

Step 4

Therefore, the solutions of the equation $|x - 6| = 5$ are as follows. (Enter your answers as a comma-separated list.)

$x =$ 11, 1 ✓

CW1.1-BB

1 **TRUE/FALSE**

The solution set of a linear equation $ax + b = 0$ with $a \neq 0$ is always $\{ -b/a \}$.

- (•) True
- ( ) False

2 **TRUE/FALSE**

An identity is an equation that is true for all real numbers x for which the terms are defined.

- (•) True
- ( ) False

3 **TRUE/FALSE**

When solving equations involving absolute values, you should consider both the positive and negative scenarios for the inner expression.

- (•) True
- ( ) False

CW1.2-BB

1 **MULTIPLE CHOICE**

Solve the quadratic equation by factoring: x^2 − 9 = 0.

- A. x = 0 or x = 9
- B. x = 9 or x = −9
- C. x = 1 or x = −1
- D. x = 3 or x = −3

2 **MULTIPLE CHOICE**

Solve the equation containing a radical: sqrt(3x + 7) = 11.

- A. x = 38
- B. x = 121/3
- C. x = 2
- D. x = 22

COURSE OUTCOME 2

CW2.1-WCR

CW2.1-WCR

100%

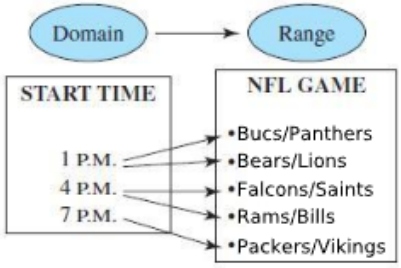
Last sync 2 weeks ago

17 questions

55 / 55 points

Correct.

Determine whether the relation is a function.



Not a function

Correct.

Determine whether this relation is a function. Assume that the coordinate pair (x, y) represents the independent variable x and the dependent variable y.

$$\{(0,3),(3,0),(7,3),(-7,3),(8,11),(-10,11)\}$$

Function

CW2.2-WCR

CW2.2-WCR

100%

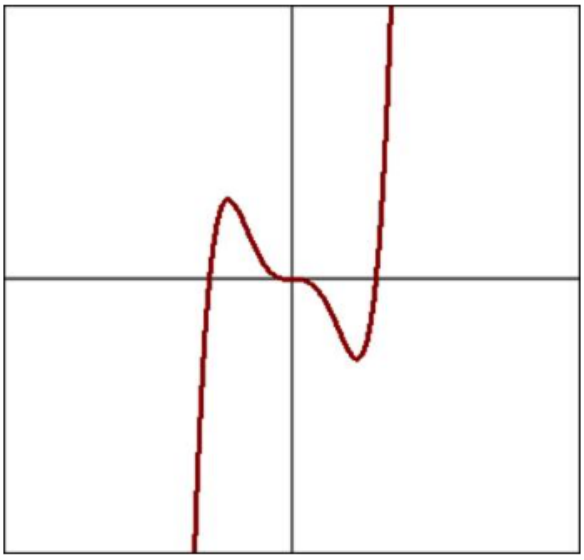
Last sync 2 weeks ago

22 questions

65 / 65 points

✔ Your answer is correct.

Determine if the function in the graph is even, odd, or neither:



- even
- odd
- neither even nor odd

Your answer is correct.

For the functions f anf g given below, find $f+g$, $f-g$, $f\cdot g$, and $\frac{f}{g}$, and the domain of each.
Enter the answers in fully simplified and expanded form.

$$f(x) = 4x + 3$$
$$g(x) = x^2 - 121$$

$f + g =$ $x^2 + 4\,x - 118$ ✓

Domain format (,) ✓ ($-\infty$, ∞) ✓ ✓

$f - g =$ $-x^2 + 4\,x + 124$ ✓

Domain format (,) ✓ ($-\infty$, ∞) ✓ ✓

$f \cdot g =$ $4\,x^3 + 3\,x^2 - 484\,x - 363$ ✓

Domain format (,) ✓ ($-\infty$, ∞) ✓ ✓

$\frac{f}{g} =$ $\frac{4\,x + 3}{x^2 - 121}$ ✓

Domain format (,) ∪ (,) ∪ (,) ✓

($-\infty$, -11) ✓ ✓ ∪ (-11 , 11) ✓ ✓ ∪ (11 , ∞) ✓ ✓

CW2.6-WCR

CW2.6-WCR

100%

Last sync 2 weeks ago

20 questions

60 / 60 points

Correct.

Please view the following video before answering this question. Lecture Video, Section 5.1, Example 02

The graph of $f(x) = b^x$ is [increasing] if $b > 1$, is [decreasing] if $b < 1$.

Correct.

Please view the following video before answering this question. Lecture Video, Section 5.1, Example 10

If $\$3{,}000$ is deposited in an account paying 3% compounded quarterly, how much will you have in the account in 5 years?

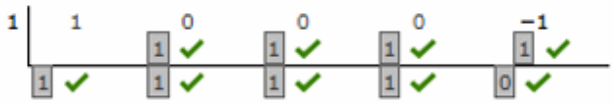
Round the answer to the nearest cent.

$ 3483.55

CW2.3-CWA

Use synthetic division to show that c is a zero of P.

$P(x) = x^4 - 1, \quad c = 1$



Verify that the given binomial is a factor of $P(x)$, and write $P(x)$ as the product of the binomial and its reduced polynomial $Q(x)$.

$P(x) = x^3 + 2x^2 + x - 4, \; x - 1$

$P(x) =$ $(x-1)(x^2+3x+4)$

CW2.4-CWA

Find all vertical asymptotes of the rational function. (Enter your answers as a comma-separated list of equations. If an answer does not exist, enter DNE.)

$$F(x) = \frac{x^2 + 11}{6x^2 - 19x - 36}$$

Find all vertical asymptotes of the rational function. (Enter your answers as a comma-separated list of equations. If an answer does not exist, enter DNE.)

$$F(x) = \frac{7x}{x^4 - 16}$$

$x = 2, x = -2$

CW2.5-CWA

Expand the given logarithmic expression. Assume all variable expressions represent positive real numbers. When possible, evaluate logarithmic expressions. Do not use a calculator.

$\log_b(xyz)$

$\log_b(x)+\log_b(y)+\log_b(z)$ ✓

Expand the given logarithmic expression. Assume all variable expressions represent positive real numbers. When possible, evaluate logarithmic expressions. Do not use a calculator.

$\ln\frac{z^3}{\sqrt{xy}}$

$3\ln z-\left(\frac{1}{2}\right)\ln(x)-\left(\frac{1}{2}\right)\ln(y)$ ✓

COURSE OUTCOME 3

CW3.1-CWA

Determine the slope of the line and the coordinates of the y-intercept.

$$f(x) = \frac{8x}{9}$$

slope 8/9 ✓

y-intercept $(x, y) = ($ 0, 0 ✓ $)$

Determine whether the graphs of the two equations are parallel, perpendicular, or neither parallel nor perpendicular.

$$f(x) = 7x - 8; \quad y = -\frac{x}{7} - 8$$

- ○ parallel
- ◉ perpendicular
- ○ neither

✓

CW3.2-CWA

Find the equation in standard form of the parabola with vertex at the origin and focus (0, 8).

$x^2 = 32y$

Find the equation in standard form of the parabola with vertex at (4, 3) and focus (4, 0).

$(x-4)^2 = -12(y-3)$

CW3.3-CWA

Find the equation in standard form of the ellipse, given the information provided.

Center $(0, 0)$, major axis of length 24, foci at $(4, 0)$ and $(-4, 0)$.

$\frac{x^2}{144} + \frac{y^2}{128} =$ ✓

Find the equation in standard form of the ellipse, given the information provided.

Major axis of length 8 on the x-axis, center at $(0, 0)$; ellipse passes through $(2, \sqrt{6})$

$\frac{x^2}{16} + \frac{y^2}{8} =$ ✓

CW3.4-CWA

Find the equation in standard form of the hyperbola that satisfies the stated conditions. (If an answer does not exist, enter DNE.)

Vertices (0, 2) and (0, −2), passing through (4, 5)

$\frac{y^2}{4} - \frac{21x^2}{64} =$ ✓

Find the equation in standard form of the hyperbola that satisfies the stated conditions. (If an answer does not exist, enter DNE.)

Vertices (−2, 2) and (6, 2), foci (−4, 2) and (8, 2)

$\frac{(x-2)^2}{16} - \frac{(y-2)^2}{20}$ ✓

CW3.5-WCR

CW3.5-WCR

100%

Syncing error. Try again.

12 questions

60 / 60 points

Your answer is correct.

State the center and radius of the equation $x^2 + y^2 + 14x - 20y + 145 = 0$.

Center: (-7 , 10)

Radius = 2